Revision plan for the Semantic Landmark Particle Filter manuscript

Target contribution

Reframe the paper around the following central claim:

This paper introduces a GNSS-aided semantic landmark particle filter that improves row-level global localisation and wrong-row recovery in repetitive vineyard environments by converting stable agricultural landmarks into row-aligned structural constraints.

Avoid claiming that the method provides fully robust or control-grade localisation in all vineyard conditions. The evidence should support a precise claim: semantic structural modelling reduces perceptual aliasing and improves recovery from adjacent-row hypotheses when GNSS is noisy but available.

Priority 1: Strengthen the experimental evidence

1. Add GNSS degradation experiments:

2. nominal noisy GNSS;

3. increased Gaussian noise;
4. drift bias;
5. multipath jump/outlier bursts;
6. short GNSS outages, for example 5 s, 10 s, 20 s;

7. headland-only degradation, since that is the main failure region.

8. Report performance as a function of GNSS quality:

9. APE;

10. cross-track error;
11. row correctness;
12. wrong-row duration;
13. recovery distance after a row error;

14. failure rate.

15. Add a parameter sensitivity analysis:

16. semantic noise;

17. GNSS noise;
18. corridor weight;
19. background weight;

20. number of particles;
21. semantic range;
22. detection confidence threshold.
23. Add detector evaluation:
24. trunk/pole precision, recall, F1, and AP/mAP;
25. qualitative examples of correct detections, missed detections, and false positives;
26. detection performance versus distance;
27. impact of detection threshold on localisation.
28. Add at least one more field sequence if possible:
29. a different day;
30. a different traversal speed;
31. denser canopy or different lighting;
32. ideally a second vineyard/orchard block.

If collecting new data is difficult, prioritise GNSS degradation, detector evaluation, and baseline correction first. These directly address the reviews.

Priority 2: Fix and strengthen the baselines

1. Re-run or verify RTAB-Map:
2. confirm camera intrinsics and extrinsics;
3. confirm frame conventions;
4. check whether the apparent wrong-direction turns in Fig. 4 are caused by misconfiguration;
5. show RTAB-Map in its own local frame as a sanity check;
6. state whether RTAB-Map is used as visual odometry, SLAM, or map-frame localisation.
7. Replace or demote RTAB-Map if it remains unsuitable:
8. present it as a visual-SLAM reference, not as a direct localisation competitor;
9. avoid using poor RTAB-Map performance as the main evidence of superiority.
10. Fully specify AMCL:
11. ROS/ROS2/Nav2 version;
12. map type;
13. laser model parameters;
14. particle count;
15. initialisation;

16. update thresholds;
17. resampling settings.
18. Fully specify AMCL+NoisyGNSS:
19. whether this is EKF/UKF/Kalman fusion or post-hoc fusion;
20. inputs and covariance values;
21. update rate;
22. whether GNSS is fused before or after AMCL;
23. why it does not improve over AMCL.
24. Add a stronger internal baseline:
25. GNSS + LiDAR particle filter without semantics;
26. GNSS + point-landmark particle filter without semantic walls;
27. semantic detections as isolated landmarks;
28. semantic walls without corridor prior;
29. semantic walls without GNSS adaptation.

This is essential because the most convincing comparison is not against RTAB-Map, but against increasingly close variants of the proposed model.

Priority 3: Clarify the method

Add a “System assumptions and initialisation” subsection covering:

- odometry source;
- odometry noise model;
- whether wheel odometry, IMU, LiDAR odometry, or RTK-derived increments are used;
- GNSS-to-map alignment;
- initial pose and heading requirements;
- coordinate frames;
- timestamp synchronisation;
- calibration between camera, LiDAR, robot base, and GNSS antenna;
- whether the semantic wall map is surveyed manually, with RTK, or generated automatically.

Add a “Likelihood design and uncertainty calibration” subsection explaining:

- which terms correspond to sensor uncertainty;
- which terms are heuristic regularisers;
- how sigma values are selected;
- how MAD normalisation affects particle weights;
- why the tempered softmax is used;
- whether parameters are tuned on the same sequences or held out.

Add a “Failure modes” subsection:

- row crossing;
- prolonged GNSS outage;
- missing landmarks;
- structural map changes;
- false semantic detections;
- headland turns;
- dense canopy or poor depth sensing.

Priority 4: Improve the metrics

Keep APE and cross-track error, but make row-level metrics more operationally meaningful.

Add:

- wrong-row duration, not only number of switches;
- recovery distance after wrong-row entry;
- percentage of time with cross-track error below thresholds such as 0.25 m, 0.5 m, 1.0 m;
- maximum consecutive wrong-row interval;
- headland-specific metrics;
- in-row versus headland breakdown.

This will directly answer the concern that row-switch event count is hard to interpret.

Priority 5: Reframe the paper narrative

The revised title and abstract should avoid overclaiming precision. A stronger title would be:

GNSS-Aided Semantic Landmark Particle Filtering for Row-Level Localisation in Vineyards

or

Semantic Wall Constraints for GNSS-Aided Vineyard Localisation under Row-Level Perceptual Aliasing

The abstract should explicitly say that the method targets row-level global consistency and wrong-row recovery, not centimetre-level navigation. The limitations should be moved earlier and stated more openly.

Priority 6: Prepare a journal-style contribution

For a journal version, expand the paper around three contributions:

1. A semantic wall representation that converts sparse stable landmarks into row-level structural constraints.
2. A GNSS-aided semantic particle-filter likelihood for resolving vineyard row aliasing.
3. A systematic field evaluation including GNSS degradation, semantic degradation, detector quality, ablations, and operational row-level metrics.

The revised paper should include an explicit response-to-review table internally during writing, mapping each reviewer criticism to a manuscript change.